

**GOOD
DESIGN**



**BAD
DESIGN**



CSE472: HCI – Human Computer Interaction

Lecture – 2: Design of Everyday Things

Spring 2026

“Create a an innovative, but cute picture for my HCI undergraduate class to show the design of everyday things what could be a good design and what could be a bad design” [ChatGPT]



Inspiring Excellence

Today: Understanding Users

- Why do we need to understand users?
- What do we need to know?
- Why do people find it difficult to understand and use so many everyday objects?
- What are Don Norman's principles and how do they apply to the design of everyday things?
- How can we apply Norman's principles to the design of computer interfaces?

We need to understand users

- Product success depends on designing systems/tools that can be used
 - Safely
 - Effectively
 - Efficiently
 - Enjoyably



We need to understand users

- Bad design results in
 - anger and frustration
 - decreased productivity in the workplace
 - higher error rates
 - physical and emotional injury
 - equipment damage
 - loss of customer loyalty
 - costs money

Why does good design matter?

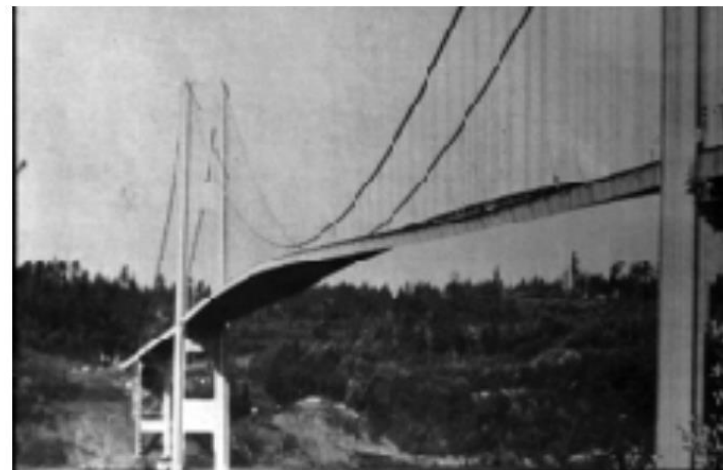
Bad artifact
design



Bad product
design



Bad
communication



Dangerous
design

Psychopathology of Everyday Things

- We are surrounded by many everyday things that have poor usability

Door knobs, stoves, showers, taps, ... hardware, software

- Many of these things can be difficult to interpret and frustrating to use if they provide no clues or false clues as to how they operate

What do we need to know about users?

- Physical & cognitive abilities (& special needs)
- Personality & culture
- Knowledge & skills
- Motivation
- Fatal Mistakes:
 - Assume all users are alike
 - Assume all users are like the designer
 - Especially YOU!

You are not the Customer



Seems obvious, but...

You have different experiences

You have different terminology

You have different ways of looking at the world

Easy to think of yourself as typical customer

Easy to make wrong assumptions

Norman's Principles of Design

- Make things visible
- Provide a good conceptual model
 - Affordances
 - Mapping
 - Constraints
 - Feedback

Visibility

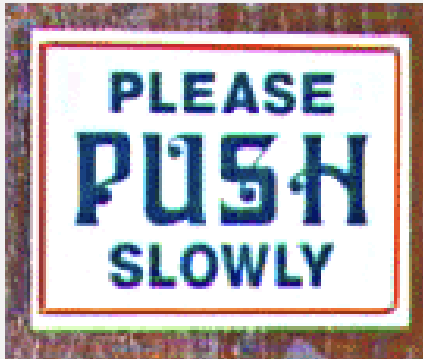
- The correct parts must be *visible* and they must convey the correct message
- Natural signals are naturally interpreted
- Visibility problems occur when clues are lacking or exist in excess
- Just by looking the user should know
 - State of the system
 - Possible actions
- Don't violate these principles to make something “look good”!

Bad Visibility: How do you use this plug?



Please Push Slowly!

- Wonder why doors are made out of glass?

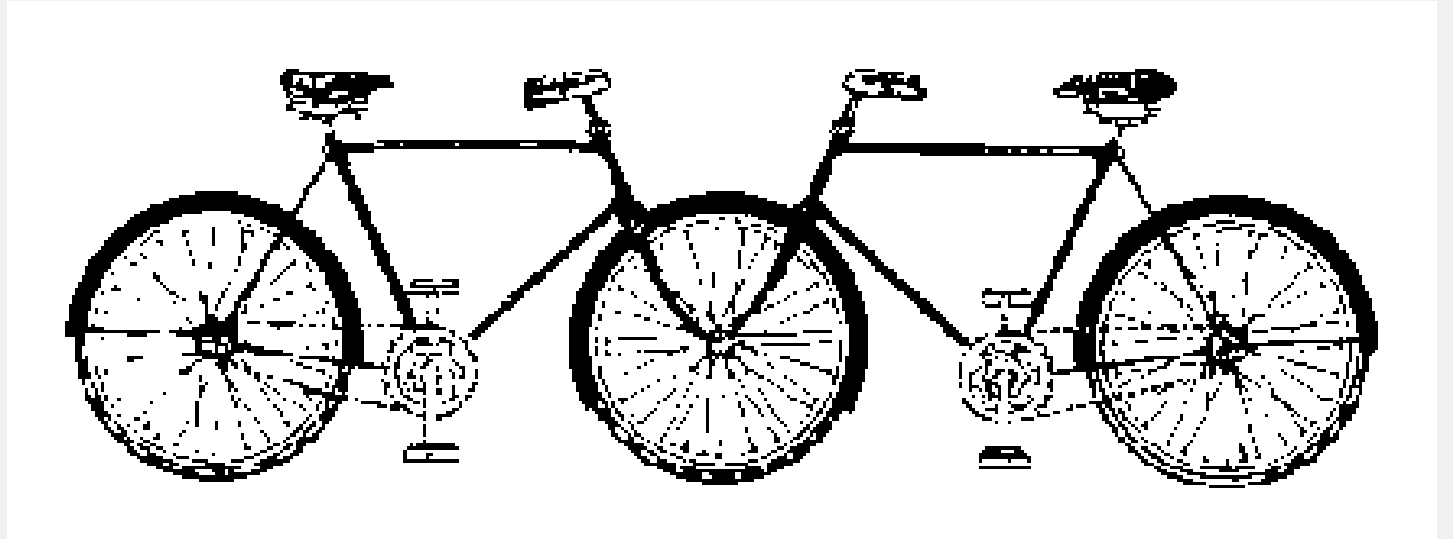


- Evidently someone was smacked...



Good Conceptual Model

- A good conceptual model allows us to predict the effects of our actions



- Without a good model we operate blindly
 - No understanding of cause or effect
 - No recourse when something breaks

Bad Conceptual Model: The Case of the Mistaken Urinal



THIS IS A MOP SINK

Affordances and Signifiers

Affordances define what actions are possible

- Button affords pushing
- Handle affords grasping
- Knob affords turning

Signifiers specify how people discover these possibilities

Signifiers are signs, perceptible signals of what can be done

Affordances and Signifiers Example



Bad Signifier: How do you open this drawer?



Affordances and Signifiers

Affordances define what actions are possible

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Signifiers specify how people discover these possibilities

Signifiers are signs, perceptible signals of what can be done

Mapping

- Controls and displays should exploit natural mapping
- Natural mapping takes advantage of physical analogies and cultural standards
 - Physical: Steering wheel
 - Cultural: red means stop, green means go
- Transfer Effects: People transfer their expectations from familiar objects to similar new ones
 - Positive transfer: previous experience applies to new situation
 - Negative transfer: previous experience conflicts with new situation

What Knob Goes Where?



Exploit Natural Mapping



Constraints

- Constraints limit the ways in which something can be used
- Constraints can be
 - Physical
 - Semantic
 - Cultural
 - Logical

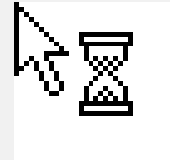


Feedback

- Feedback is sending back to the user information about what action has actually been done
- Visibility of the effects of the operation tell you if something worked correctly
- Systems should be designed to provide adequate feedback to the users to ensure they know what to do next in their tasks

Feedback Examples

- Phone button press tones
- Rice cooker goes “bing!”
- Clicker on your turn signal
- Animated icon while waiting for a web page to load



Norman's Principles of Design

- Make things visible
- Provide a good conceptual model
 - Affordances
 - Mapping
 - Constraints
 - Feedback

What about software??

What about Software?

- Visibility

Visibility of the tasks the interface supports
Communication of system state / mode

- Affordance

If it looks like a button it can be pressed,
if it is underlined it can be clicked (web)

- Mapping

Clicking on a particular interface element produces expected effect (e.g., Open should be under File)

What about Software?

- Constraints

Constraining search criteria, graying out menu items that don't apply in a particular context

- Feedback

Providing clear and immediate feedback for each action
Animated icons for any waiting

- Many more examples of these that we will see during the semester...

Larson's dog effect



Same with Software

Thank you for registering! We appreciate your business. To activate your software, you will be sent an email key. After you have received the key by email you will be able to [click here](#) and proceed with the activation.

Blah blah blah blah blah blah blah blah blah blah blah
blah blah blah blah blah blah blah blah blah blah blah
blah blah blah blah blah blah blah blah blah [click here](#) blah
blah blah blah blah blah.

How do we learn about users?

- If there are usability problems in everyday “simple” things, the challenge is much greater for complex software
- Problems can be overcome through HCI methods that collect and analyze data to better understand your users
- There are many methods, we will learn and practice a few....
 - Contextual enquiry
 - Surveys
 - Interviews
 - Usage Analysis

How do we learn about users?



Contextual Enquiry

Observe users in their natural environment while they perform real tasks. Discover problems they can't articulate in interviews.



Surveys

Gather quantitative data from many users simultaneously. Ideal for measuring attitudes, satisfaction, and demographic patterns.



Interviews

In-depth one-on-one conversations reveal user goals, frustrations, and mental models in ways surveys cannot.



Usage Analysis

Analyse logs, heatmaps and analytics to understand how users actually behave — not just what they say they do.

Key Takeaways

- 1 Good design is invisible – users accomplish their goals without thinking about the interface.
- 2 Bad design costs real money: lost productivity, errors, injuries, and customer churn.
- 3 You are NOT your user – designers must research and observe real users.
- 4 Norman's 6 principles (Visibility, Conceptual Model, Affordances, Mapping, Constraints, Feedback) are the foundation of usable design.
- 5 These principles apply equally to physical products and digital software interfaces.



Any questions?